Data Analysis Project

Superstore Sales Data Analysis

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Tools Used:

Python

Pandas

Matplotlib

Seaborn

Year: 2026

1. Project Overview

This project analyzes a retail sales dataset to understand business performance across different product categories, regions, and customer segments.

The goal of this analysis is to identify sales trends, top-performing products, and factors affecting profitability.

Through this project, we explore patterns in sales, profit, discounts, and regional performance to generate meaningful business insights.

2. Project Objective

The main objectives of this project are:

- Analyze total sales across product categories
- Identify the best-performing regions
- Find the top-selling products
- Understand the relationship between sales and profit
- Analyze the impact of discounts on profitability

This analysis helps businesses make data-driven decisions to improve revenue and profit.

3. Dataset Description

The dataset contains retail sales data from a global superstore. Important columns in the dataset include:

- order_id – Unique identifier for each order
- order_date – Date when the order was placed
- ship_date – Shipping date of the order
- ship_mode – Shipping method used
- customer_name – Name of the customer
- segment – Customer segment (Consumer, Corporate, Home Office)
- state – Customer state
- country – Customer country
- market – Market region
- region – Sales region
- product_id – Unique product ID
- category – Product category
- sub_category – Product subcategory
- product_name – Name of the product
- sales – Total sales value
- quantity – Quantity sold
- discount – Discount applied
- profit – Profit generated
- shipping_cost – Cost of shipping
- order_priority – Priority level of the order
- year – Year of the order

4. Data Loading and Exploration

The dataset was loaded using pandas and explored to understand its structure.

```
import pandas as pd

data = pd.read_csv("SuperStoreOrders.csv")

data.head()
data.info()
data.columns
```

5. Data Cleaning

Data cleaning ensures that the dataset is ready for analysis.

```
data.isnull().sum()
```

If missing values exist:

```
data = data.dropna()
data.duplicated().sum()
```

Duplicate rows were also checked.

6. Category Wise Sales Analysis

This analysis identifies which product category generates the highest revenue.

```
category_sales = data.groupby("category")["sales"].sum()

print(category_sales)
```

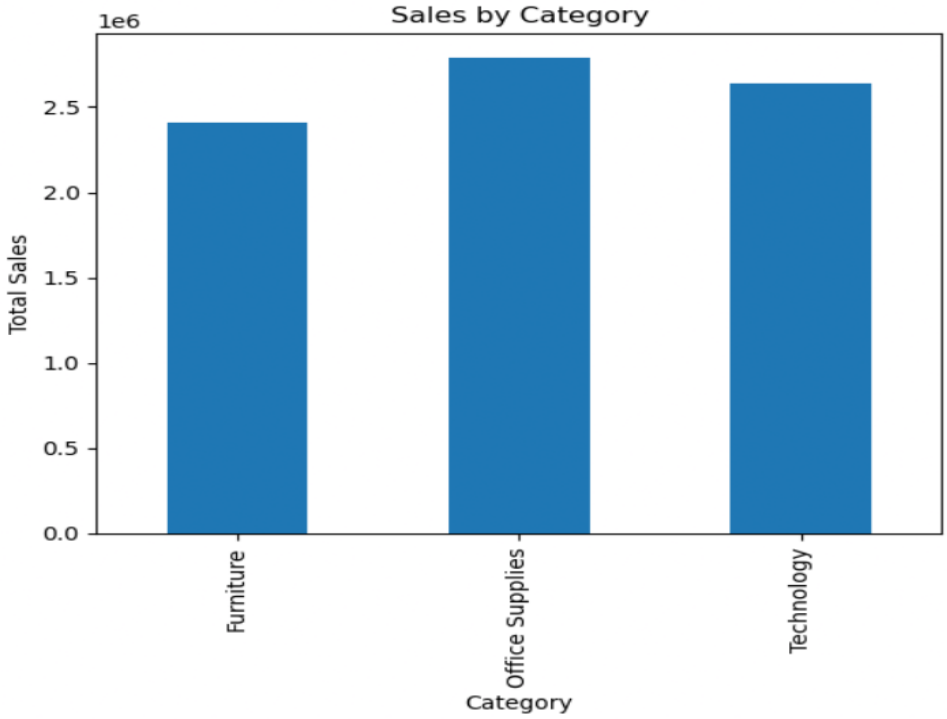
Visualization:

```
import matplotlib.pyplot as plt

category_sales.plot(kind="bar")

plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
```

plt.show()



7. Region Wise Sales Analysis

This analysis determines which region performs best in terms of sales.

region_sales = data.groupby("region")["sales"].sum()

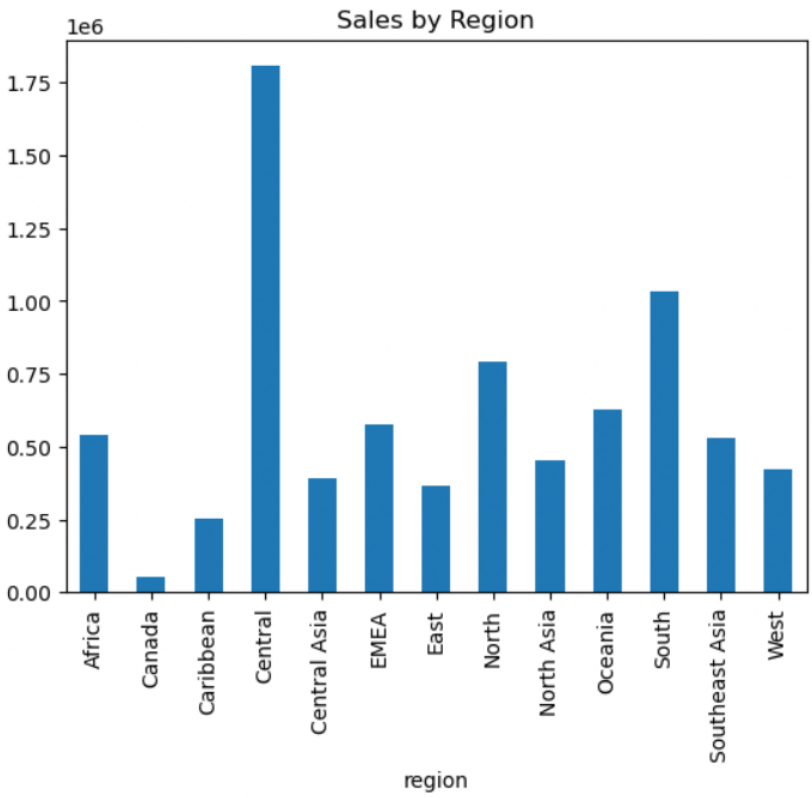
print(region_sales)

```
region
Africa              538115.0
Canada               50314.0
Caribbean           251441.0
Central            1806638.0
Central Asia        389506.0
EMEA                575562.0
East                366492.0
North               790546.0
North Asia          453686.0
Oceania             625382.0
South              1031101.0
Southeast Asia      532172.0
West                424173.0
Name: sales, dtype: float64
```

Visualization:
region_sales.plot(kind="bar")

plt.title("Sales by Region")

plt.show()

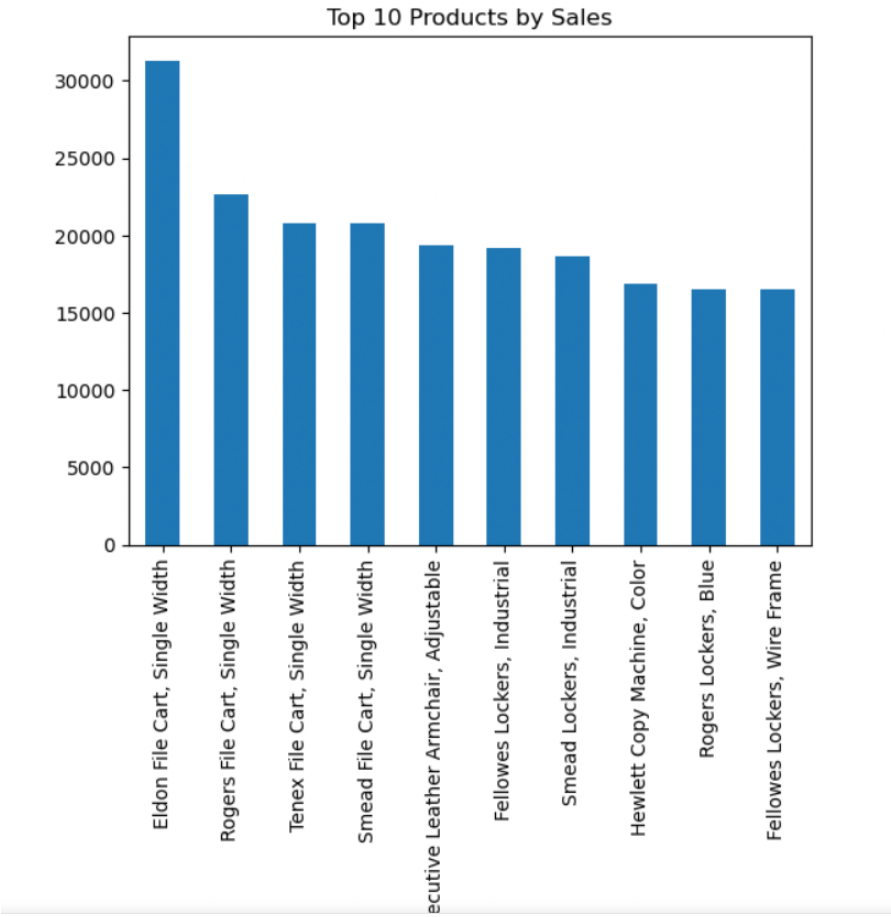


8. Top Selling Products

This analysis identifies the top 10 products generating the highest sales.

```
top_products =
data.groupby("product_name")["sales"].sum().sort_values(ascending=False)
.head(10)

print(top_products)
```



9. Sales vs Profit Analysis

This analysis examines the relationship between sales and profit.

```
import seaborn as sns

sns.scatterplot(x="sales", y="profit", data=data)

plt.title("Profit vs Sales")

plt.show()
```

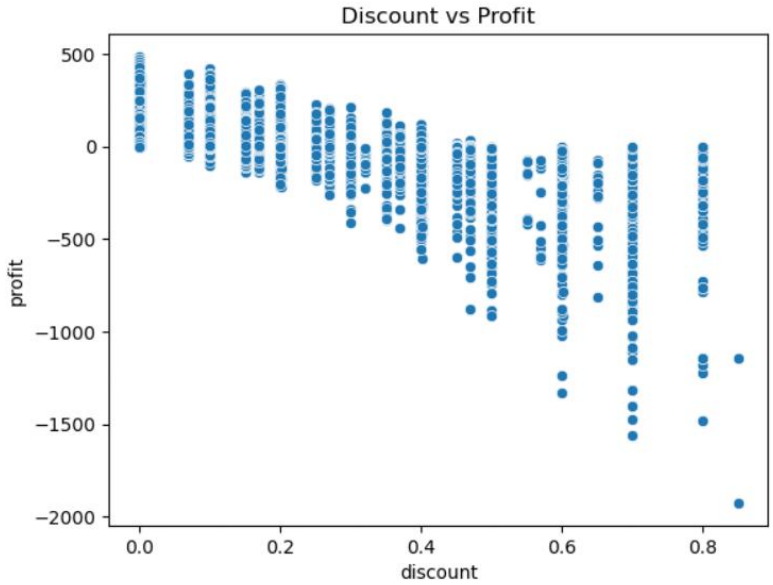
10. Discount Impact on Profit

This analysis explores how discounts affect profit.

sns.scatterplot(x="discount", y="profit", data=data)

plt.title("Discount vs Profit")

plt.show()



11. Segment Wise Sales

This analysis shows how different customer segments contribute to total sales.

```
segment_sales = data.groupby("segment")["sales"].sum()

print(segment_sales)
```

```
segment
Consumer       4058118.0
Corporate      2369261.0
Home Office    1407749.0
Name: sales, dtype: float64
```

The Consumer segment contributes the highest share of total sales compared to Corporate and Home Office segments.

12. Key Business Insights

Based on the analysis, the following insights were observed:

- **The Technology category generates the highest revenue among all product categories.**
- **The West region shows the highest overall sales performance.**
- **Some products generate high sales but relatively low profit margins.**
- **Higher discounts tend to reduce overall profitability.**
- **Customer segments contribute differently to total revenue.**

These insights help businesses optimize pricing strategies, product focus, and regional marketing efforts.

13. Conclusion

This project demonstrates how data analysis can be used to uncover valuable business insights from sales data.
By analyzing sales trends, product performance, and profit patterns, businesses can make informed decisions to improve revenue and profitability.
The analysis highlights the importance of managing discounts, focusing on high-performing product categories, and optimizing regional sales strategies.